MTH1001

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2021

Mathematical Structures

Module Leader: Prof Nigel Byott

Duration: 2 HOURS (+30 minutes upload time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN NOTE** examination

SECTION A

1. (a) Using the extended Euclidean Algorithm, calculate $\gcd(157, 61)$ and express it in the form $157a + 61b$ for suitable integers a, b. Hence solve the congruence $61x \equiv 5 \pmod{157}$. (12)

   (b) Let $\sigma = (123)(567)$ and $\tau = (12345)(67)$ in the symmetric group S_7. Find $\sigma\tau$, $\tau\sigma$, τ^2 and $\sigma\tau\sigma^{-1}$, writing each as a product of disjoint cycles. Find the order and the sign of each of these permutations. (13)

   (c) Consider the following vectors in $\mathbb{R}^4$:

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 2 \\ -2 \\ -2 \\ 4 \end{pmatrix}, \quad \mathbf{w}_3 = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{w}_4 = \begin{pmatrix} 2 \\ -4 \\ 3 \\ 3 \end{pmatrix},$$

$$\mathbf{b} = \begin{pmatrix} 4 \\ -8 \\ -5 \\ 5 \end{pmatrix}.$$

   Find a maximal linearly independent subset of $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$. Also, determine whether $\mathbf{b}$ can be written as a linear combination of $\mathbf{w}_1$, $\mathbf{w}_2$, $\mathbf{w}_3$, $\mathbf{w}_4$ and, if so, write $\mathbf{b}$ in this way. (15)

   (d) Let $\phi : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map, and let $\psi : \mathbb{R}^n \to \mathbb{R}^n$ be the function defined by $\psi(\mathbf{v}) = 2\mathbf{v} - \phi(\mathbf{v})$ for all $\mathbf{v} \in \mathbb{R}^n$. Show that ψ is also a linear map. (10)

[**50**]

SECTION B

2. (a) Let n be a positive integer and let a, $b \in \mathbb{Z}$. Give the definition of $a \equiv b \pmod{n}$. (2)

   (b) Determine whether each of the following statements is true or false, and give a proof (if it is true) or a counterexample (if it is false). In each case n, a and b are positive integers.

   (i) if $a \equiv b \pmod{n}$ then $a^2 \equiv b^2 \pmod{n}$;
   (ii) if $a \equiv b \pmod{n}$ then $2^a \equiv 2^b \pmod{n}$;
   (iii) if $2a \equiv 2b \pmod{n}$ then $a \equiv b \pmod{n}$;
   (iv) if $a \equiv b \pmod{n}$ then $3a \equiv 3b \pmod{3n}$. (11)

   (c) Let G and H be groups, both written multiplicatively, and let e_G and e_H be their identity elements.

   (i) State what it means for a function $\phi : G \to H$ to be a homomorphism.
   (ii) Show that if ϕ is a homomorphism then $\phi(e_G) = e_H$.
   (iii) Let $\phi : G \to H$ be an injective homomorphism and let $g \in G$ have (finite) order n. Show that the element $\phi(g)$ of H also has order n. (12)

   [25]

3. (a) Show that $\sqrt{45}$ is irrational.

   You may use without proof the fact that every positive integer has a unique prime factorisation (up to the order of the factors), but if you do so, you should indicate clearly where you use this fact. (10)

   (b) Let $\mathbf{Fn}(\mathbb{R})$ be the real vector space of all functions $f : \mathbb{R} \to \mathbb{R}$. For each of the following subsets of $\mathbf{Fn}(\mathbb{R})$, determine whether the subset is a *subspace* of $\mathbf{Fn}(\mathbb{R})$. Justify your answers.

   (i) $V_1 = \{f \in \mathbf{Fn}(\mathbb{R}) : f(-x) = f(x) \text{ for all } x \in \mathbb{R}\}$;
   (ii) $V_2 = \{f \in \mathbf{Fn}(\mathbb{R}) : f(2n) = -3f(n) \text{ for all } n \in \mathbb{Z}\}$;
   (iii) $V_3 = \{f \in \mathbf{Fn}(\mathbb{R}) : f(n) = n \text{ for all } n \in \mathbb{Z}\}$;
   (iv) $V_4 = \{f \in \mathbf{Fn}(\mathbb{R}) : f(x) \in \mathbb{Z} \text{ for all } x \in \mathbb{R}\}$. (15)

   [25]

4. (a) Let V be a vector space over the real numbers $\mathbb{R}$, and let U, W be two subspaces of V. Let $\mathbf{u}_1, \ldots, \mathbf{u}_r$ be a basis of U, and let $\mathbf{w}_1, \ldots, \mathbf{w}_s$ be a basis of W. Show that if $\mathbf{u}_1, \ldots, \mathbf{u}_r, \mathbf{w}_1, \ldots, \mathbf{w}_s$ are linearly independent then $U \cap W = \{\mathbf{0}\}$. (10)

   (b) Let $\mathbb{Q}^\times$ be the group of all non-zero rational numbers (with the operation of multiplication). Which of the following subsets of $\mathbb{Q}^\times$ are subgroups? Justify your answers.

   (i) $\{x \in \mathbb{Q}^\times : x > 0\}$;
   (ii) $\{x \in \mathbb{Q}^\times : x > -1\}$;
   (iii) $\{1\}$;
   (iv) $\{x \in \mathbb{Q}^\times : x = a/b \text{ with } a \equiv b \equiv 1 \pmod 3\}$;
   (v) $\{x \in \mathbb{Q}^\times : x = p/q \text{ where both } p \text{ and } q \text{ are prime numbers}\}$. (15)

[**25**]